

The subcaption package^{*}

Axel Sommerfeldt

<https://gitlab.com/axelsommerfeldt/caption>

2023/02/19

Abstract

This package offers an user interface to typeset sub-captions.

- ▲ At the end of each section, text marked with the mountain symbol will contain background knowledge on how the particular command or environment is actually implemented. If you just want to use this package as it is, you don't have to read or understand them.
- ▲ Since version 3.1 the `caption` package offers a low-level interface to typeset sub-captions: `\DeclareCaptionSubType` defines the required counters and internal commands, `\setcaptionsubtype` switches to the sub-caption mode, and `\caption@subtypehook` could be extended to apply own code when a switch to the sub-caption mode is in progress. This package demonstrates its usage by offering a high-level user interface additionally.

^{*}This package has version number v1.6.

Contents

1	Loading the package	3
2	Setting options	3
3	The <code>\subcaptionbox</code> command	4
3.1	Comparison with <code>\captionbox</code>	7
4	The <code>subcaptionblock</code> environment	7
5	The <code>subcaptiongroup</code> environment	9
6	The <code>\DeclareCaptionSubType</code> command	11
7	Cross Referencing	12
7.1	The <code>\subref</code> command	13
7.2	The <code>subrefformat=</code> option	13
7.3	Referencing sub-figures without sub-captions	14
7.4	Typesetting sub-captions without generating a (new) reference	15
7.5	Where to place the <code>\label</code> command?	16
7.6	Where do hyperlinks jump?	16
8	Captions inside sub-figures	17
8.1	Using the <code>overpic</code> package	17
8.2	Using the <code>stackengine</code> package	18
8.3	Using the <code>tikz</code> package	19
8.4	Using the <code>skip</code> and <code>margin</code> options	21
9	Numbering	21
9.1	Pitfall #1: Using multiple main captions	21
9.2	Pitfall #2: Expecting <code>\caption</code> to increment the counter	23
10	Abbreviatory commands	24
11	The <code>\subfloat</code> command	25
12	Required packages	26
13	Other packages with similar offerings	26
14	Beyond this package	27
15	Thanks	27

1 Loading the package

Load this package using

```
\usepackage[<options>]{subcaption} .
```

The options for the subcaption package are the same ones as for the caption package, but specify settings which are used for sub-captions *additionally*. In fact

```
\usepackage[<options>]{subcaption}
```

is identical to

```
\usepackage{subcaption}  
\subcaptionsetup{<options>}
```

The default settings for sub-captions are:

```
margin=0pt,font+=smaller,labelformat=parens,labelsep=space,  
skip=6pt,list=false,hypcap=false1
```

2 Setting options

`\subcaptionsetup` The `\subcaptionsetup` command sets options specifically for sub-captions.

New feature
v1.6

```
\subcaptionsetup{<options>}
```

sets options valid for all sub-captions.

An example:

```
\subcaptionsetup{font=it}
```

sets the font to “italic” for all sub-captions.

```
\subcaptionsetup[<environment>]{<options>}
```

is an alternative syntax to

```
\captionsetup[sub<environment>]{<options>}
```

¹This means that sub-captions are not listed in the List of Figures or Tables by default, but you can enable it by specifying the option `list=true`.

An example:

```
\subcaptionsetup[table]{labelformat=simple}
```

is identical to

```
\captionsetup[subtable]{labelformat=simple}
```

and sets the `labelformat` to “simple” for all sub-captions inside tables.

Options specified with `\usepackage[...]{subcaption}` and `\subcaptionsetup{...}` will override the ones specified by `\captionsetup{...}` and `\captionsetup[figure]{...}`, but are again overwritten by `\subcaptionsetup[figure]{...}` (same for ‘table’). So finally we have the following order how settings for sub-captions are applied:

1. Global settings (`\usepackage[...]{caption}` and `\captionsetup{...}`)
2. Environmental settings (`\captionsetup[figure -or- table]{...}`)
3. Local settings (`\captionsetup{...}` inside figure or table environment)
4. Default ‘sub’ settings (`margin=0pt, font+=smaller, ...`, see above)
5. Custom ‘sub’ settings (`\usepackage[...]{subcaption}` and `\subcaptionsetup{...}`)
6. Environmental ‘sub’ settings (`\subcaptionsetup[figure -or- table]{...}`)
7. Local ‘sub’ settings (`\captionsetup{...}` inside subfigure or subtable)

An example:

```
\usepackage[labelsep=quad,indentation=10pt]{caption}  
\usepackage[labelfont=bf,list=true]{subcaption}  
\captionsetup[table]{textfont=it,position=top}  
\subcaptionsetup[table]{textfont=sf}
```

causes sub-captions inside table environments to be typeset with the settings

```
indentation=10pt,position=top,margin=0pt,font=small,  
labelformat=pars, labelsep=space, skip=6pt,hyperc=false,  
labelfont=bf,list=true,textfont=sf .
```

3 The `\subcaptionbox` command

`\subcaptionbox` The `\subcaptionbox` command typesets given content and caption. It automatically aligns the sub-figures resp. sub-tables by their very first caption line.

Its syntax is:

```
\subcaptionbox[⟨list entry⟩]{⟨heading⟩}[⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}
\subcaptionbox*{⟨heading⟩}[⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}
```

The arguments $\langle list\ entry \rangle$ & $\langle heading \rangle$ will be used for typesetting the $\backslash caption$.

$\langle width \rangle$ is the width of the resulting $\backslash parbox$; the default value is the width of the contents.

$\langle inner-pos \rangle$ specifies how the contents will be justified inside the resulting $\backslash parbox$; it can be either ‘c’ (centered, using $\backslash centering$), ‘l’ (left-justified, using $\backslash raggedright$), ‘r’ (right-justified, using $\backslash raggedleft$), or ‘s’ (for no special justification). The default is ‘c’. (But you can use any justification defined with $\backslash DeclareCaptionJustification$ as well, e.g.: ‘centerlast’)

When using $\backslash subcaptionbox$, the baseline of the resulting box will be placed right between contents and heading. So usually you don’t have to care about the vertical alignment of the sub-figures for yourself. Also the hyperlink anchor is placed properly with respect to the $hypcap=$ setting.

An example:

```
\begin{figure}
  \centering
  \subcaptionbox{A cat\label{cat}}
    {\includegraphics{cat}}
  \subcaptionbox{An elephant\label{elephant}}
    {\includegraphics{elephant}}
  \caption{Two animals}\label{animals}
\end{figure}
```

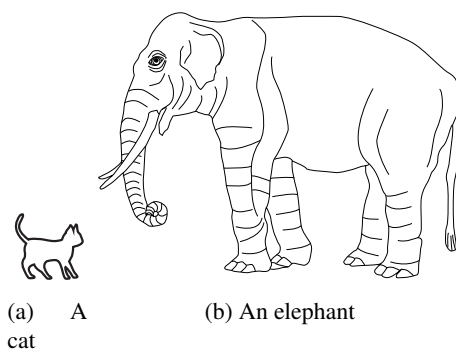


Figure 1: Two animals²

As you see the result is not satisfying; the caption below the cat looks ugly because of the small width of the graphic. This can be solved by using the optional arguments of $\backslash subcaptionbox$, increasing the width of the resulting box:

²The pictures were taken with permission from the L^AT_EX Companion[1] examples.

```
...
\subcaptionbox{A cat\label{cat}}
[2.5cm]{\includegraphics{cat}}
...
```

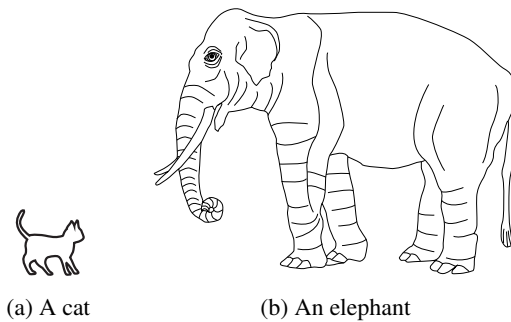


Figure 2: Two animals

Furthermore the main caption, which is centered with respect to the `\textwidth`, looks mis-aligned with respect to the sub-captions. This can (again) be solved by using the optional arguments of `\subcaptionbox`, giving both boxes the same width, for example:

```
...
\subcaptionbox{A cat\label{cat}}
[.4\textwidth]{\includegraphics{cat}}%
\subcaptionbox{An elephant\label{elephant}}
[.4\textwidth]{\includegraphics{elephant}}
...
```

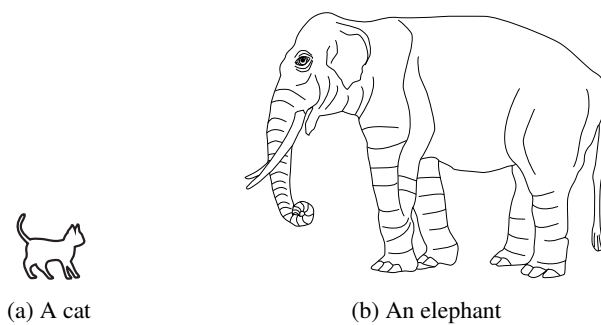


Figure 3: Two animals

▲ The `\subcaptionbox` is a `\parbox` with `\setcaptionsubtype` as first contents line.

3.1 Comparison with `\captionbox`

Both, `\captionbox` (offered by the `caption` package) and `\subcaptionbox`, put its contents and caption into a `\parbox` of either natural or given width and share the same (mandatory and optional) arguments, but while `\captionbox` uses a regular caption, `\subcaptionbox` uses a sub-caption instead, like “(a)” or “2.1”.

So for example the last example would look like this when using `\captionbox` instead of `\subcaptionbox`:

```
\begin{figure}
  \centering
  \captionbox{A cat\label{cat}}{
    [.4\textwidth]{\includegraphics{cat}}%
  }
  \captionbox{An elephant\label{elephant}}{
    [.4\textwidth]{\includegraphics{elephant}}
  }
\end{figure}
```



Figure 4: A cat

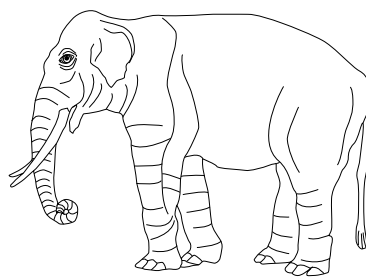


Figure 5: An elephant

4 The `subcaptionblock` environment

`subcaptionblock` (*env.*) The `subcaptionblock` environment makes a box with given width. Inside this box the regular caption commands (like `\caption`, `\phantomcaption`, ...) could be used to typeset sub-captions.

New feature
v1.5

`subcaptionblock` has the same (optional & mandatory) arguments as the `minipage` environment:

```
\begin{subcaptionblock} [<outer-pos>] [<height>] [<inner-pos>] {<width>}
...
\end{subcaptionblock}
```

The default value for *<outer-pos>* is either ‘b’ (if no main caption was typeset so far) or ‘t’ (if a main caption was already typeset), and the default value for *<inner-pos>* is ‘s’.

New feature
v1.2

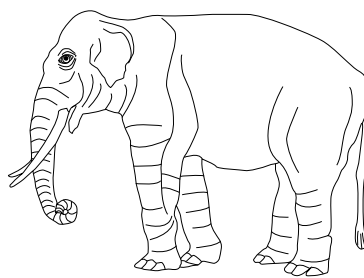
Beside the *<outer-pos>* values of ‘c’, ‘t’, and ‘b’, the `subcaption` package also offers the values ‘T’ and ‘B’ additionally which align the contents at the very top resp. bottom. (In contrast ‘t’ and ‘b’ align the contents at the top resp. bottom *baseline*.)

The same example as [Figure 3](#), but this time using the `subcaptionblock` environment instead of `\subcaptionbox`:

```
\begin{figure}
\centering
\begin{subcaptionblock}{.4\textwidth}
\centering
\includegraphics{cat}
\caption{A cat}\label{cat}
\end{subcaptionblock}%
\begin{subcaptionblock}{.4\textwidth}
\centering
\includegraphics{elephant}
\caption{An elephant}\label{elephant}
\end{subcaptionblock}%
\caption{Two animals}\label{animals}
\end{figure}
```



(a) A cat



(b) An elephant

Figure 6: Two animals

Some additional notes:

- You can override the settings for a specific subcaption with a `\captionsetup` inside the `subcaptionblock`, e.g.:

```
\begin{subcaptionblock}{.4\textwidth}
\centering
\includegraphics{owl}
\captionsetup{skip=3pt}
\caption{An owl}\label{owl}
\end{subcaptionblock}
```

- Just like `figure` or `table`, a `subcaptionblock` could have multiple captions, e.g.:

```
\begin{subcaptionblock}{.4\textwidth}
\centering
\includegraphics{cat}
\caption{A cat}\label{cat}
```

```

\medskip
\includegraphics{elephant}
\caption{An elephant}\label{elephant}
\end{subcaptionblock}

```

- Hyperlinks targeted to this sub-figure will jump to the beginning of the subcaptionblock, and not to the `\caption` inside it (if `hypcap=true` is set for sub-captions). (See 7.6)

`subfigure (env.)` The `subcaptionblock` environment is also offered as `subfigure` resp. `subtable (env.)` table. (And prior version 1.5 of the `subcaption` package it was only available as `subfigure` resp. `subtable`.)

There is no difference in them except the environment name should match the current floating environment, i.e. inside a figure a `subfigure` should be used, and inside a table a `subtable` should be used. Using the wrong sub-environment will cause a warning since v1.5 of the `subcaption` package.

So if in doubt, or when writing own $\LaTeX$ commands which should work in every floating environment, using `subcaptionblock` is the correct choice.

(Note that the default value for $\langle outer-pos \rangle$ has changed from ‘c’ to either ‘b’ or ‘t’ in version 1.5 of the `subcaption` package.)

▲ The `subcaptionblock`, `subfigure`, and `subtable` environments are minipage environments with `\setcaptionsubtype` as first contents line.

`subfigure` and `subtable` are defined with the help of `\ForEachCaptionSubType` offered by the `caption` package, which executes code for every sub-type declared with `\DeclareCaptionSubType`.

5 The subcaptiongroup environment

`subcaptiongroup (env.)` The `subcaptiongroup` environment is only switching to the sub-caption mode inside an own $\TeX$ group. Inside this environment the regular caption commands (like `\caption`, `\phantomcaption`, ...) could be used to typeset sub-captions.

New feature
v1.5

Its syntax is:

```

\begin{subcaptiongroup}
...
\end{subcaptiongroup}

```

There is a starred variant of this environment as well which uses `\setcaptionsubtype*` instead of `\setcaptionsubtype` internally:

```

\begin{subcaptiongroup*}
...
\end{subcaptiongroup*}

```

While this gives you great flexibility, it also offers you no help formatting its contents.

The same example as [Figure 6](#), but this time using the `subcaptiongroup` environment instead of `\subcaptionblock`:

```
\begin{figure}
  \centering
  \begin{subcaptiongroup}
    \centering
    \parbox[b]{.4\textwidth}{%
      \centering
      \includegraphics{cat}
      \caption{A cat}\label{cat}}%
    \parbox[b]{.4\textwidth}{%
      \centering
      \includegraphics{elephant}
      \caption{An elephant}\label{elephant}}%
  \end{subcaptiongroup}
  \caption{Two animals}\label{animals}
\end{figure}
```

–or–

```
\begin{figure}
  \centering
  \begin{subcaptiongroup}
    \centering
    \begin{minipage}[b]{.4\textwidth}
      \centering
      \includegraphics{cat}
      \caption{A cat}\label{cat}
    \end{minipage}%
    \begin{minipage}[b]{.4\textwidth}
      \centering
      \includegraphics{elephant}
      \caption{An elephant}\label{elephant}
    \end{minipage}
  \end{subcaptiongroup}
  \caption{Two animals}\label{animals}
\end{figure}
```

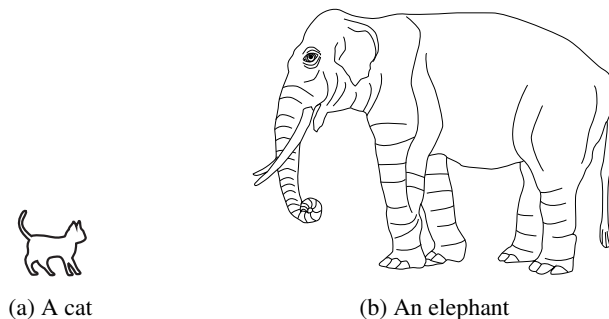


Figure 7: Two animals

▲ The `subcaptiongroup` environment is a $\text{\LaTeX}$ environment with `\setcaptionsubtype` as first contents line.

6 The `\DeclareCaptionSubType` command

`\DeclareCaptionSubType` For using the sub-caption feature of the caption package some commands and counters must be prepared. This is done with

```
\DeclareCaptionSubType[⟨numbering scheme⟩]{⟨type⟩}
\DeclareCaptionSubType*[⟨numbering scheme⟩]{⟨type⟩}
```

For the environments `figure` & `table`, and all the ones defined with `\DeclareFloatingEnvironment` offered by the `newfloat` package, this will be done automatically, but for other environments (e.g. the ones defined with `\newfloat` offered by the `float` package or `\DeclareNewFloatType` offered by the `floatrow` package) this has to be done manually.

The starred variant provides the sub-caption numbering format `⟨type⟩.⟨subtype⟩` (for example ‘1.2’) while the non-starred variant simply uses `⟨subtype⟩` (for example ‘a’).

Own numbering formats can be created by redefining `\thesub⟨type⟩`, e.g.:

```
\DeclareCaptionSubType*{figure}
\renewcommand\thesubfigure{\thefigure\alph{subfigure}}
```

would give you sub-caption numbers like ‘1b’.

The default numbering scheme is `alph`, but you can use any $\text{\LaTeX}$ (or self-defined) command name here which converts a counter to a text value, e.g.: `arabic`, `roman`, `Roman`, `alph`, `Alph`, `fnsymbol`, ...

But `\DeclareCaptionSubType` is not only for defining new sub-caption types, you can use this command for re-definitions as well, e.g.:

```
\DeclareCaptionSubType*[arabic]{table}
\subcaptionsetup[table]{labelformat=simple,labelsep=colon}
```

will give you sub-captions in tables like these ones:

Table 1: Two tables			
1.1: Table one		1.2: Table two	
A	B	E	F
C	D	G	H

▲ `\DeclareCaptionSubType` and `\ForEachCaptionSubType` are integral parts of the caption package kernel.

7 Cross Referencing

The macro `\the<counter>` is not only responsible for the look of the `<counter>`, but for the look of the references typeset with `\ref`, too. References will be prefixed by `\LaTeX` with the internal macro `\p@<counter>`.

`\DeclareCaptionSubType` will define both of them for sub-captions (e.g. subfigure and subtable), and as you have seen in the last section `\DeclareCaptionSubType` will give you some options to control the internal (re-)definition of `\the<counter>` and `\p@<counter>`.

`\thesubfigure` For example `\thesubfigure` and `\p@subfigure` are (as default) internally defined as

```
\newcommand\thesubfigure{\alph{subfigure}}
\newcommand\p@subfigure{\thefigure}
```

so the label of sub-captions will look like ‘a’ (decorated by the selected label format), while references will look like ‘1a’ since they are prefixed by `\p@subfigure = \thefigure`.

After `\DeclareCaptionSubType*[arabic]{figure}`, `\thesubfigure` and `\p@subfigure` will look like

```
\renewcommand\thesubfigure{\thefigure.\arabic{subfigure}}
\renewcommand\p@subfigure{ }
```

But if you want detailed control on how the references will look like, the options of `\DeclareCaptionSubType` are potentially not sufficient. In this case one need to redefine these two macros on his/her own. Some examples:

If you want parentheses around the sub-figure part of the reference, so they will look like ‘1 (a)’, you may get them this way:

```
\usepackage[labelformat=simple]{subcaption}
\renewcommand\thesubfigure{ (\alph{subfigure}) }
```

(Note: Since `parens` is the default label format you will get double parentheses in sub-captions when not specifying a different label format, e.g. `simple`.)

But if you want only a closing parenthesis, so references will look like ‘1a)’, but the sub-captions itself should still look like ‘(a)’, this would be a possible solution:

```

\usepackage{subcaption}
\renewcommand\thesubfigure{\alph{subfigure}}
\DeclareCaptionLabelFormat{opening}{(#2)}
\subcaptionsetup[figure]{labelformat=opening}

```

 Please note that you need to surround redefinitions of `\p@⟨counter⟩` with `\makeatletter` and `\makeatother`. See <http://tex.stackexchange.com/questions/8351/> for details.

7.1 The `\subref` command

While `\ref{⟨key⟩}` (and `\ref*{⟨key⟩}`, if the `hyperref` package is used) usually gives a combined result representing the main caption counter and the sub-caption one, it is sometimes useful to have a reference to the sub-caption only. For this purpose you can use

```

\subref{⟨key⟩}
\subref*{⟨key⟩} 3 .

```

So for example `\ref{cat}` gives the result ‘1a’ but `\subref{cat}` gives ‘a’.

Note: If the sub-caption was (re-)defined with the starred variant `\DeclareCaptionSubType*`, both `\ref` and `\subref` usually gives the same result.

 The `\subref` command demonstrates the usage of `\caption@subtypehook` which will be called during `\captionsetup{subtype}`.

7.2 The `subrefformat=` option

`subrefformat=` By applying `\DeclareCaptionSubType`, or by redefining `\the⟨counter⟩` and `\p@⟨counter⟩`, you will change the look of references typeset with `\ref` and `\subref`.

New feature

v1.1 But maybe you only want to change the output of `\subref` without affecting the references typeset with `\ref`?

This is possible, too, by using the option `subrefformat`:

```

\captionsetup{subrefformat=⟨label format⟩}

```

This one will choose a label format (either a pre-defined one, or a one defined with `\DeclareCaptionLabelFormat`) as decorative element to sub-references. The default one is `simple` which has no decorative elements but simply typeset the reference as it is.

For example

```

\captionsetup{subrefformat=parens}

```

will result in references (typeset with `\ref`) like ‘1a’ but sub-references (typeset with `\subref`) like ‘(a)’.

7.3 Referencing sub-figures without sub-captions

`\phantomcaption` If you don't want to give a sub-figure a caption (yet), because the picture itself already contains the caption, or for some other reason, you could use the command

```
\phantomcaption
```

instead of `\caption`.

`\phantomcaption` is offered by the `caption` package since version 3.2 and does not generate any output but increases the sub-figure resp. sub-table counter and gives you an anchor for a `\label` command which can be placed after it.

An example:

```
\begin{figure}
  \centering
  \begin{subcaptiongroup}
    \includegraphics{cat_with_a}
    \phantomcaption\label{cat}
    \includegraphics{elephant_with_b}
    \phantomcaption\label{elephant}
  \end{subcaptiongroup}
  \captionsetup{subrefformat=parens}
  \caption{Two animals: \subref{cat} a cat,
           and \subref{elephant} an elephant}
  \label{animals}
\end{figure}
```

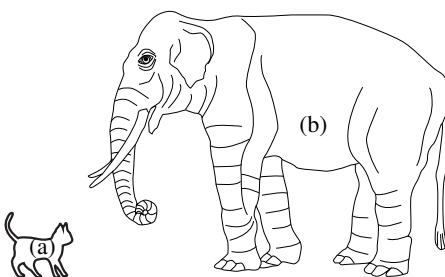


Figure 8: Two animals: (a) a cat, and (b) an elephant

`\captionlistentry` If you don't want to give a sub-figure a caption (yet), because the picture itself already contains the caption, or for some other reason, you could also use the command

```
\captionlistentry{\list entry}
```

instead of `\caption`.

`\captionlistentry` is offered by the `caption` package since version 3.3 and (just like `\phantomcaption`) does not generate any output but increases the sub-figure resp. sub-table counter and gives you an anchor for a `\label` command which can be placed after it. Additionally to `\phantomcaption` this command puts an entry into the list of figures resp. tables.

³Like `\ref*`, `\subref*` is only available if the `hyperref` package^[4] is used.

An example:

```
\begin{figure}
\centering
\begin{subcaptiongroup}
\includegraphics{cat_with_a}
\captionlistentry{A cat}
\label{cat}
\includegraphics{elephant_with_b}
\captionlistentry{An elephant}
\label{elephant}
\end{subcaptionblock}
\captionsetup{subrefformat=parens}
\caption{Two animals: \subref{cat} a cat,
        and \subref{elephant} an elephant}
\label{animals}
\end{figure}
```

7.4 Typesetting sub-captions without generating a (new) reference

The `\caption` command is a multi-purpose command:

1. It increments the sub-figure resp. sub-table counter and generates an internal reference which could be used with `\label`
2. It puts an entry into the list of figures resp. tables
3. It finally typesets a caption

When put into a command or into an environment which either evaluates its content more than once or does not like one of the first two actions (for whatever reason), the result could be either an error message or an incorrect result, for example a sub-figure resp. sub-table counter which was incremented more than once.

`\captiontext` In these cases the `\caption` command could be split into `\phantomcaption` which performs step one only (or `\captionlistentry` which performs steps one and two), and `\captiontext` which performs step three only. This way critical steps could be out-sourced from the target command or environment, for example by prepending `\phantomcaption` and using `\captiontext` inside.

The syntax of `captiontext` is

```
\captiontext [<number>] {<text of sub-caption>}
\captiontext* [<number>] {<text of sub-caption>}
```

`\captiontext` is offered by the `caption` package since version 3.6 and as opposite to `\caption` it does not increase the sub-figure resp. sub-table counter and does not give you an anchor for a `\label` command. It typesets the caption only, using existing counter values unless a *<number>* is given explicitly.

For example code please take a look at [8](#).

7.5 Where to place the `\label` command?

When `\caption` inside a `captionsubblock`, `subfigure`, `subtable`, or `caption-subgroup` environment, the `\label` can be either placed inside the caption text or right after the command, e.g.:

```
\caption{Some text here\label{text}}
...
\caption{Some other text}\label{othersetext}
...
\caption{Something completely different}
\label{differenttext}
```

When using `\phantomcaption` or `\captionlistentry` inside a `captionsubblock`, `subfigure`, `subtable`, or `captionsubgroup` environment, the `\label` should be placed right after the command, e.g.:

```
\phantomcaption\label{this}
...
\phantomcaption
\label{that}
```

But when using the `\subcaptionbox` command, the `\label` should be placed inside the caption text, e.g.:

```
\subcaptionbox{A description here\label{todo1}}
{Some content here}
...
\subcaptionbox[List-of-Figures entry]
{A description here\label{todo2}}
{Some content here}
```

Placing `\label` outside the `\subcaptionbox` would produce an incorrect reference.

7.6 Where do hyperlinks jump?

For the `captionsubblock`, `subfigure`, `subtable`, and `captionsubgroup` environments, and for the `\subcaptionbox` command (and all other constructs which use `\setcaptionsubtype`) the hyperlink anchors will be placed in respect to the `hypcap=` setting. While usage of this option is straight-forward for ordinary captions, the usage for sub-captions depends on the setting regarding the main captions.

This table gives you an overview where the hyperlinks will jump:

caption subcaption	hypcap=false	hypcap=true
	sub-caption	figure or table (default setting)
hypcap=false		
hypcap=true	sub-figure or sub-table	sub-figure or sub-table

But if the `captionsubgroup*` environment is used (or a different construct which uses `\setcaptionsubtype*`) and `hypcap=true` is set for sub-captions, the sub-caption package does not know where the sub-figure or sub-table actually begins, so it will jump to the sub-caption instead.

Remember: If you use the `hypcap` package[5], it controls the placement of the hyperlink anchors, making the rules above invalid.

(See also the documentation of the `caption` package, sections about `hyperref` & `hypcap`.)

8 Captions inside sub-figures

Nearly all code examples so far have placed the caption either above or below the sub-figure. But it is possible to put the caption inside the sub-figure, too:

- The caption could already be part of the image. (This case is already handled in 7.3.)
- $\text{\LaTeX}$ packages like `overpic`, `stackengine`, or `tikz` could be used. Note that the commands resp. environments offered by these packages usually evaluate their content more than once, resulting in either errors or wrong reference counters. For this reason `\caption` should not be used here, instead `\phantomcaption` or `\captionlistentry` should be used outside the command resp. environment and `\captiontext` should be used inside it. See also 7.4
- The options `skip=` and `margin=` could be used to place the caption onto the image.

8.1 Using the `overpic` package

The same example as in section 7.3, but using the `overpic` environment offered by the `overpic` package [7] to place the captions inside the pictures:

```
\usepackage{overpic}
...
\begin{figure}
  \centering
  \begin{subcaptiongroup}
    \subcaptionlistentry{A cat}
    \label{cat}
    \begin{overpic}[width=60pt]{cat}
      \put(40,34){\captiontext*{}}
    \end{overpic}
    \subcaptionlistentry{An elephant}
    \label{elephant}
    \begin{overpic}[width=140pt]{elephant}
      \put(58,40){\captiontext*{}}
    \end{overpic}
  \end{subcaptiongroup}
\end{figure}
```

```

\captionsetup{subrefformat=parens}
\caption{Two animals: \subref{cat} a huge cat,
        and \subref{elephant} an elephant}
\end{figure}

```

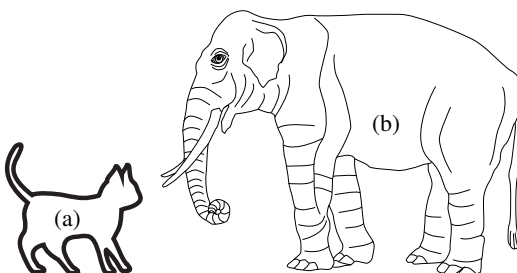


Figure 9: Two animals: (a) a huge cat, and (b) an elephant

If neither a reference to the sub-figures nor an entry in the List of Figures is needed, the usage of `\phantomcaption` resp. `\captionlistentry` could be dropped. Since this leaves `\captiontext` without a valid sub-figure number value it must be given to it explicitly as optional argument.

Furthermore the usage of the `subcaptiongroup` environment could be dropped here, instead it would be sufficient to replace `\captiontext` with `\subcaptiontext`. (See 10)

```

\usepackage{overpic}
...
\begin{figure}
  \centering
  \begin{overpic}[width=60pt]{cat}
    \put(40,34){\subcaptiontext*[1]{} }
  \end{overpic}
  \begin{overpic}[width=140pt]{elephant}
    \put(58,40){\subcaptiontext*[2]{} }
  \end{overpic}
  \caption{Two animals: A huge cat and an elephant}
\end{figure}

```

8.2 Using the stackengine package

The same example as in section 7.3, but using the `\stackinset` command offered by the `stackengine` package [8] to place the captions inside the pictures:

```

\usepackage{stackengine}
...
\begin{figure}
  \centering

```

```

\begin{subcaptiongroup}
  \subcaptionlistentry{A cat}
  \label{cat}
  \stackinset{1}{25pt}{b}{20pt}{\captiontext*{}}
    {\includegraphics[width=60pt]{cat}}
  \subcaptionlistentry{An elephant}
  \label{elephant}
  \stackinset{1}{80pt}{b}{60pt}{\captiontext*{}}
    {\includegraphics[width=140pt]{elephant}}
\end{subcaptiongroup}
\captionsetup{subrefformat=parens}
\caption{Two animals: \subref{cat} a huge cat,
        and \subref{elephant} an elephant}
\end{figure}

```

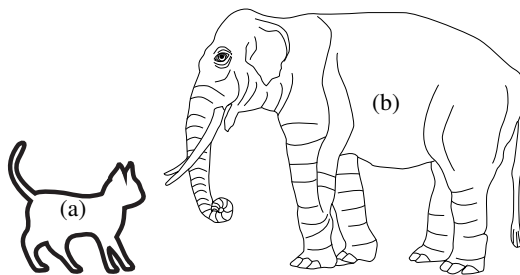


Figure 10: Two animals: (a) a huge cat, and (b) an elephant

If neither a reference to the sub-figures nor an entry in the List of Figures is needed, this code could be simplified to:

```

\usepackage{stackengine}
...
\begin{figure}
  \centering
  \stackinset{1}{25pt}{b}{20pt}{\subcaptiontext*[1]{}}
    {\includegraphics[width=60pt]{cat}}
  \stackinset{1}{80pt}{b}{60pt}{\subcaptiontext*[2]{}}
    {\includegraphics[width=140pt]{elephant}}
  \caption{Two animals: A huge cat and an elephant}
\end{figure}

```

8.3 Using the tikz package

The same example as in section 7.3, but using the `tikzpicture` environment offered by the `tikz` package [9] to place the captions inside the pictures:

```

\usepackage{tikz}
...

```

```

\begin{figure}
  \centering
  \begin{subcaptiongroup}
    \subcaptionlistentry{A cat}
    \label{cat}
    \begin{tikzpicture}
      \node (cat) at (0,0)
        {\includegraphics[width=60pt]{cat}};
      \node at (0.1,-0.1) {\captiontext*{}};
    \end{tikzpicture}
    \subcaptionlistentry{An elephant}
    \label{elephant}
    \begin{tikzpicture}
      \node (elephant) at (0,0)
        {\includegraphics[width=140pt]{elephant}};
      \node at (0.5,-0.1) {\captiontext*{}};
    \end{tikzpicture}
  \end{subcaptiongroup}
  \captionsetup{subrefformat=parens}
  \caption{Two animals: \subref{cat} a huge cat,
    and \subref{elephant} an elephant}
\end{figure}

```

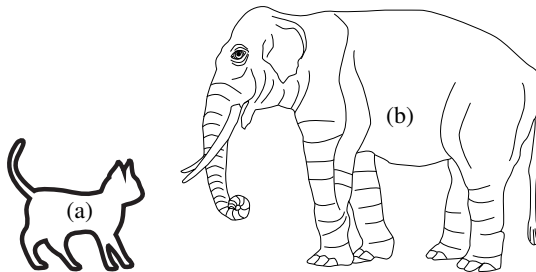


Figure 11: Two animals: (a) a huge cat, and (b) an elephant

If neither a reference to the sub-figures nor an entry in the List of Figures is needed, this code could be simplified to:

```

\usepackage{tikz}
...
\begin{figure}
  \centering
  \begin{tikzpicture}
    \node (cat) at (0,0)
      {\includegraphics[width=60pt]{cat}};
    \node at (0.1,-0.1) {\subcaptiontext*[1]{}};
  \end{tikzpicture}
  \begin{tikzpicture}

```

```

\begin{tikzpicture}
\node (elephant) at (0,0)
{\includegraphics[width=140pt]{elephant}};
\node at (0.5,-0.1) {\subcaptiontext*[2]{} };
\end{tikzpicture}
\caption{Two animals: A huge cat and an elephant}
\end{figure}

```

8.4 Using the skip and margin options

For a particular sub-caption the `skip=...` could be set to a negative value so it will overlap with the image. Combined with `singlelinecheck=off` (to switch off the centering of short captions) and `margin=...` it could be placed at a specific horizontal position within the image, too.

The same example as in section 7.3, but using the `skip=...` and `margin=...` options:

```

\begin{figure}
\centering
\subcaptionsetup[figure]
{skip=-28pt,slc=off,margin={25pt,0pt}}
\subcaptionbox{\label{cat}}
{\includegraphics[width=60pt]{cat}}
\subcaptionsetup[figure]
{skip=-60pt,slc=off,margin={80pt,0pt}}
\subcaptionbox{\label{elephant}}
{\includegraphics[width=140pt]{elephant}}
\captionsetup{subrefformat=parens}
\caption{Two animals: \subref{cat} a huge cat,
and \subref{elephant} an elephant}
\end{figure}

```

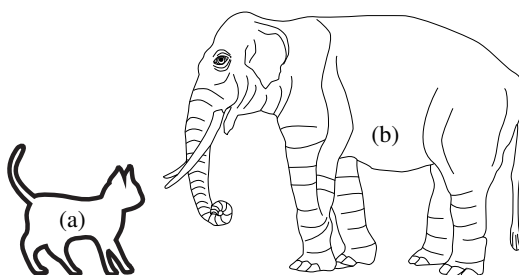


Figure 12: Two animals: (a) a huge cat, and (b) an elephant

9 Numbering

9.1 Pitfall #1: Using multiple main captions

When multiple main captions are used within a figure or table, and sub-captions are used as well, how does the subcaption package know which sub-captions belong to which

caption, i.e. what is the main counter value for the sub-captions?

Let's illustrate this problem with an example document:

```
\documentclass{article}
\usepackage{graphicx, subcaption}
\begin{document}
\begin{figure}
  \centering
  \includegraphics[width=4cm]{example-image-c}
  \caption{Caption no. 1}
  \bigskip
  \subcaptionbox
    {\label{fig:2a}}
    {\includegraphics[width=2cm]{example-image-a}}
  \subcaptionbox
    {\label{fig:2b}}
    {\includegraphics[width=2cm]{example-image-b}}
  \caption{Caption no. 2}
\end{figure}
Look at sub-figures \ref{fig:2a} and \ref{fig:2b}.
\end{document}
```

It's obvious that the sub-captions belong to 2nd caption, and therefore `\ref{fig:2a}` will become "2a", isn't it? But since the subcaption package is only involved when using $\LaTeX$ commands either defined or patched by the caption or subcaption package, this is what the caption package is aware of:

```
\usepackage{subcaption}
\begin{document}
\begin{figure}
  \caption{Caption no. 1}
  \subcaptionbox
    {\label{fig:2a}}
    {\langle unknown content \rangle}
  \subcaptionbox
    {\label{fig:2b}}
    {\langle unknown content \rangle}
  \caption{Caption no. 2}
\end{figure}
\end{document}
```

So from captions point of view it's not easy to decide if the sub-captions belong to the 1st or 2nd main caption since they are placed between them. (Note: The subcaption package is only offering an user interface to the sub-caption feature of the caption package, and therefore this decision is the responsibility of the caption package.)

But how does the caption package makes a decision? If in doubt, it rather clings to the past than to the future, i.e. in this case it decides that the sub-captions belong to the 1st caption and therefore the result of `\ref{fig:2a}` is not "2a" but "1a". If the caption package is unsure about its decision (like in this case), a warning will be issued:

Package caption Warning: Ambiguous sub-caption(s),
 use \nextfloat on input line 15.
 See the caption package documentation for explanation.

How to fix it? Use `\nextfloat` to tell the caption package where the 2nd figure within the figure environment starts:

```
\documentclass{article}
\usepackage{graphicx,subcaption}
\begin{document}
\begin{figure}
  \centering
  \includegraphics[width=4cm]{example-image-c}
  \caption{Caption no. 1}
  \bigskip
  \nextfloat
  \subcaptionbox
    {\label{fig:2a}}
    {\includegraphics[width=2cm]{example-image-a}}
  \subcaptionbox
    {\label{fig:2b}}
    {\includegraphics[width=2cm]{example-image-b}}
  \caption{Caption no. 2}
\end{figure}
Look at sub-figures \ref{fig:2a} and \ref{fig:2b}.
\end{document}
```

 Note that the `\nextfloat` command was introduced in caption package v3.6. Previous versions of the caption package have bound the decision to the `position=` setting of the floating environment instead, i.e. sub-captions belonged to the caption above if `position=top` was set and they belonged to the caption below if `position=below` was set. While this would be beneficial in this case, there were several cases where it was not and especially it was not always comprehensible to the user why his references got an incorrect numbering.

Furthermore the new decision algorithm always succeeds if there is only one caption within the figure or table (which is the case most of the time) while the old one did not.

If you still prefer the old decision algorithm (for example because you want to process an already existing document), you need to specify the caption package version explicitly, for example:

```
\documentclass{article}
\usepackage{caption}[=v3.5]
\usepackage{graphicx,subcaption}
...
```

See also: [12](#)

9.2 Pitfall #2: Expecting `\caption` to increment the counter

Usually `\caption` increments the figure resp. table counter and therefore it is usually safe to assume that the counter was not incremented yet in code used before `\caption`:

```
\documentclass{article}
\begin{document}
\begin{figure}
  Figure counter before caption: \thefigure
  \caption{Some text}
  Figure counter after caption: \thefigure
\end{figure}
\end{document}
```

Unsurprisingly the results are 0 and 1 for the counter values.

But this happens if we use the subcaption package:

```
\documentclass{article}
\usepackage{subcaption}
\begin{document}
\begin{figure}
  \subcaptionbox{}{some content}
  \subcaptionbox{}{some content}
  Figure counter before caption: \thefigure
  \caption{Some text}
  Figure counter after caption: \thefigure
\end{figure}
\end{document}
```

Here the results are 1 and 1 for the counter values. Why?

Since the caption package assumes that there will be a `\caption` following `\subcaptionbox` it decides that the sub-captions belong to the upcoming main caption and therefore share the same main counter value which needs to be incremented before its use. Therefore the first `\subcaptionbox` in the figure increments the figure counter while the second one and especially the `\caption` does not.

Usually this is no problem at all but keep this in mind if you are doing tricky stuff with the figure resp. table counter within figures resp. tables.

10 Abbreviatory commands

<code>\subcaption</code> <code>\phantomsubcaption</code> <code>\subcaptionlistentry</code> <code>\subcaptiontext</code>	As we have seen in sections 8.1, 8.2, and 8.3 it's sometimes inconvenient to use the commands and environments described so far. For example it would be inconvenient to use <code>\captiongroup</code> just for a single <code>\captiontext</code> in this particular case:
--	---

```
...
\begin{overpic}[width=60pt]{cat}
  \put(40,34)
  {\begin{captiongroup*}
    \captiontext*[1]{}
```

```

\end{captiongroup*}}
\end{overpic}
...

```

Since `\captiontext` is already used in an extra environment (`overpic`) we don't need an extra `captiongroup*` here, instead using `\setcaptionsubtype*` (which switches into the sub-caption mode without making an `hyperref` anchor) would be sufficient:

```

...
\begin{overpic}[width=60pt]{cat}
  \put(40,34)
    {\setcaptionsubtype*
     \captiontext*[1]{} }
\end{overpic}
...

```

But this is still inconvenient when used many times, and therefore the `subcaption` package defines several extra commands which are prefixed with `\setcaptionsubtype*`:

Regular command	using <code>\setcaptionsubtype*</code>	available since
<code>\caption</code>	<code>\subcaption</code>	v1.0
<code>\phantomcaption</code>	<code>\phantomsubcaption</code>	v1.1
<code>\captionlistentry</code>	<code>\subcaptionlistentry</code>	v1.5
<code>\captiontext</code>	<code>\subcaptiontext</code>	v1.5

This way our code snippet above could be simplified to:

```

...
\begin{overpic}[width=60pt]{cat}
  \put(40,34){\subcaptiontext*[1]{} }
\end{overpic}
...

```

 Since `\setcaptionsubtype` should only be used within an extra group or environment, the same applies to these commands as well. (In this case `\subcaptiontext` is encapsulated by the `overpic` environment, so we are ok here.

 The `\subcaption` command is just a simple combination of `\setcaptionsubtype*` and `\caption`. Same for all other abbreviatory commands here.

11 The `\subfloat` command

`\subfloat` To allow a smoother transition from the `subfig` package [10] (which is unmaintained for over 16 years) this package also offers `\subfloat` with the same syntax:

New feature v1.3

```

\subfloat[\list_entry][\sub-caption]{\body}

```

12 Required packages

New feature v1.4 Starting with version 1.4 the subcaption package requires at least version 3.1 of the caption package and loads it automatically. (Older versions of the subcaption package have required exactly the version of the caption package which was released with it.)

If you need to use a specific version of the caption package you need to load it *before* the subcaption package, e.g.:

```
\usepackage[...]{caption}[=v3.5]
\usepackage[...]{subcaption}
```

Note that there are limitations if an older version of the caption package is used:

- The `\phantomsubcaption` command need at least caption v3.2.
- The `\subcaptionlistentry` command need at least caption v3.3.
- The `\subcaptiontext` command need at least caption v3.6.
- The `\subfloat` emulation needs at least caption v3.4.

13 Other packages with similar offerings

The subcaption package is not the only one defining sub-figure counters and offering commands or environments to typeset sub-figures. The most established are:

- The memoir document class
- The subfigure package and its successor, the subfig package

New feature v1.6 Prior subcaption version 1.6 an error message was issued if the ‘subfigure’ or ‘subtable’ counter was already defined by any of them. Since subcaption version 1.6 the ‘subfigure’ and ‘subtable’ counters will only be defined by the subcaption package if they are not defined yet, but the usage of any of the commands or environments described in this document will result in an error message “The counter sub... was defined by ...” if the corresponding counter was not defined by the subcaption package. Equally the commands or environments offered by the above document classes or packages should not be used with counters defined by the subcaption package, this will result in either L^AT_EX errors or unwanted behaviour.

So the rule of thumb is: If the counter ‘subfigure’ or ‘subtable’ is defined by document class or package xyz, then the corresponding commands and environments offered by the document class or package xyz should be used.

An example document:

```
\documentclass{memoir}
\newsubfloat{table}      % defines the 'subtable' counter
\usepackage{subcaption} % defines the 'subfigure' counter
\begin{document}
\begin{figure}
\subcaptionbox{...}{...} % <- This is fine
\caption{...}
```

```

\end{figure}
\begin{table}
\caption{...}
\subcaptionbox{...}{...} % <- This results in an error,
\end{table}               \subtop should be used instead
\end{document}

```

14 Beyond this package

For a more advanced usage of the sub-caption feature of the caption package, please take a look at the excellent keyfloat package[6] which provides the environments keysubfigs, keysubtabs, and keysubfloats for typesetting sub-figures and sub-tables. Furthermore the floatrow package[3] provides the subfloatrow environment for typesetting sub-figures.

15 Thanks

I would like to thank Stephen Dalton who helped to make this package a better one.

References

- [1] Frank Mittelbach and Michel Goossens:
The L^AT_EX Companion (2nd. Ed.),
Addison-Wesley, 2004.
- [2] Axel Sommerfeldt:
Customizing captions of floating environments,
2022/01/07
- [3] Olga Lapko:
The floatrow package documentation,
2007/12/24
- [4] Sebastian Rahtz & Heiko Oberdiek:
Hypertext marks in L^AT_EX,
November 12, 2007
- [5] Heiko Oberdiek:
The hypcap package – Adjusting anchors of captions,
2007/04/09
- [6] Brian Duun:
The L^AT_EX keyfloat Package,
2019/09/23
- [7] Rolf Niepraschk:
The overpic package,
2020/02/22

- [8] Steven B. Segletes:
The stackengine Package,
July 22, 2021
- [9] Till Tantau:
The TikZ and PGF Packages,
May 15, 2021
- [10] Steven D. Cochran:
The subfig package,
2005/07/05